

8. 此方程有实根 $\Leftrightarrow \phi_0 \geq 0 \Leftrightarrow k_{22}$

$$\therefore P(k \geq 2) = \frac{3}{5}$$

$$10. P(X > 10) = \int_{10}^{\infty} 5e^{-\frac{x}{5}} dx = e^{-2}$$

$$Y \sim BC(5, e^{-2}) \quad \therefore EY = 5e^{-2}$$

$$P(Y=1) = 1 - P(Y=0) - P(Y=2) \\ = 0.5167$$

$$14. \bar{X} = \frac{X-\mu}{\sigma}$$

$$2. E|X| = 2 \int_0^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx = \sqrt{\frac{2}{\pi}}$$

$$\therefore E|X-\mu| = \sigma \sqrt{\frac{2}{\pi}}$$

15. 记 m 为中位数.

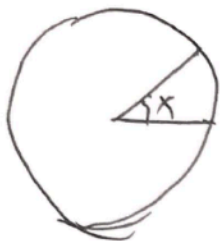
$$2. E|X-m| = \int_{-\infty}^m (m-x)f(x)dx + \int_m^{\infty} (x-m)f(x)dx \\ = -\int_{-\infty}^m xf(x)dx + \int_m^{\infty} xf(x)dx$$

$$\text{若 } a < m, \text{ 则 } E|X-a| = \int_{-\infty}^m (a-x)f(x)dx - \int_a^m (a-x)f(x)dx + \int_m^{\infty} (x-a)f(x)dx - \int_a^m (x-a)f(x)dx \\ = \int_{-\infty}^{\infty} xf(x)dx - \int_{-\infty}^m xf(x)dx + 2 \int_a^m (x-a)f(x)dx \\ \geq E|X-m|$$

若 $a < m$ 时, 同理有 $E|X-a| \geq E|X-m|$

$$\therefore E|X-m| = \min_{a \in \mathbb{R}} E|X-a|$$

16.



$$X \sim U(0, 2\pi)$$

$$L = 2R \sin \frac{X}{2}$$

$$\begin{aligned} E[L] &= \int_0^{2\pi} 2R \sin \frac{x}{2} \cdot \frac{1}{2\pi} \cdot dx \\ &= \frac{4R}{\pi} \end{aligned}$$

22. 古程有实招 $\Leftrightarrow Y \leq X^2$

~~而 $X \in (0, 1)$ 时~~

$$f(x, y) = f_X(x) f_Y(y)$$

$$\therefore P = \iint_{\substack{Y \leq X^2, X \in (0, 1), \\ Y > 0}} f(x, y) dx dy = \frac{1}{e}$$

23. ~~$f_X(x) = \int_{-\infty}^{+\infty} f(x, y) dy = \begin{cases}$~~

$$\text{Q } f_X(1) = \int_{-\infty}^{+\infty} f(1, y) dy = 0 \quad f_Y(1) = \int_{-\infty}^{+\infty} f(x, 1) dx = \frac{1}{2}$$

$$f(1, 1) = \frac{1}{2} \neq f_X(1) f_Y(1) \quad \therefore \text{不独立}$$

$$25. \quad (1) \quad E X = \int_{-\infty}^{+\infty} \frac{x}{2} e^{-|x|} dx = 0$$

$$E X^2 = \int_{-\infty}^{+\infty} \frac{x^2}{2} e^{-|x|} dx = 2 \int_0^{+\infty} \frac{x^2}{2} e^{-x} dx = 2$$

$$D X = E X^2 - (E X)^2 = 2$$

$$(2) \quad E(X|X|) = \int_{-\infty}^{+\infty} x|x| \frac{e^{-|x|}}{2} dx = 0$$

$$\therefore \text{Cov}(X, |X|) = E(X|X|) - E X E|X| = 0$$

($\therefore$ 不相关)

(3) 又 $\forall a > 0$,

$$P(|X| < a, X < a) = P(|X| < a) \neq P(|X| < a) P(X < a)$$

$\therefore$ 不独立

$$30. \quad E X = \int_{-\pi}^{\pi} \sin \theta \frac{1}{2\pi} d\theta = 0$$

$$E X Y = \int_{-\pi}^{\pi} \sin \theta \cos \theta \frac{1}{2\pi} d\theta = 0 \quad \therefore \text{Cov}(X, Y) = E X Y - E X E Y = 0$$

$\therefore X, Y$ 不相关,

$$P(X > \frac{\sqrt{2}}{2}, Y > \frac{\sqrt{2}}{2}) = 0$$

$$P(X > \frac{\sqrt{2}}{2}) = \frac{1}{4} \quad P(Y > \frac{\sqrt{2}}{2}) = \frac{1}{4}$$

$$P(X > \frac{\sqrt{2}}{2}, Y > \frac{\sqrt{2}}{2}) \neq P(X > \frac{\sqrt{2}}{2}) P(Y > \frac{\sqrt{2}}{2})$$

$\therefore$ 不独立

$$34 \quad f_{x|y}(x|y) = \frac{f(x,y)}{f_Y(y)} = \begin{cases} \frac{\lambda e^{-\lambda y}}{\frac{1}{\lambda} e^{-\lambda y}} & x > y \\ 0 & x \leq y \end{cases}$$

$$E(X|Y=y) = \int_{-y}^{+\infty} x \frac{\lambda e^{-\lambda(x-y)}}{\lambda} dx = y + \frac{1}{\lambda}$$

$$E(X^2|Y=y) = \int_{-y}^{+\infty} x^2 \frac{\lambda e^{-\lambda(x-y)}}{\lambda} dx = y^2 + \frac{2y}{\lambda} + \frac{2}{\lambda^2}$$

$$\therefore \text{Var}(X|Y=y) = E(X^2|Y=y) - (E(X|Y=y))^2 = \frac{1}{\lambda^2}$$